University of St.Gallen

Course and Examination Fact Sheet: Autumn Semester 2025

7,043: Mobile Sensing and Behavioral Metrics

ECTS credits: 6

Overview examination/s

(binding regulations see below)
decentral - Presentation, Analog, Group work group grade (25%)
Examination time: Term time
decentral - Written examination, Digital, Individual work individual grade (50%, 90 mins.)
Examination time: Term time
decentral - Written work, Digital, Group work group grade (25%)
Examination time: Term time

Attached courses

Timetable -- Language -- Lecturer
7,043,1.00 Mobile Sensing and Behavioral Metrics -- English -- Stachl Clemens

Course information

Course prerequisites

General interest in human behavior and how smartphones and mobile sensors can be used to measure how people behave across different contexts.

This course is assigned to the profile "Technology Solution Architect" and "Start-up & Scale-up Entrepreneurship", but can also be taken without selecting a specialisation.

Learning objectives

After taking this course you should:

- be able to explain, describe, and differentiate methods to measure human behavior outside of the laboratory. Know about methodological characteristics and be able to name advantages and disadvantages of each method.
- you will be able to describe judge the unique advantages of novel, objective forms of data collection (i.e., mobile sensing).
- be able to list behaviors and situational factors that can be measured with mobile sensing methods in comparison with other methods (e.g., surveys)
- be able to judge behavioral metrics regarding their suitability for specific business applications and come up with examples of their effective use
- to assess and debate the risks and opportunities of data-rich applications and to critically reflect on sensing practices from an ethical and privacy perspective

Course content

The advent of smartphones and other consumer electronics has drastically changed human life and our societies. Activities such as communication, shopping, gaming, or dating increasingly happen digitally, online, and on-the-go. The range of sensors and logging routines, built into modern consumer electronics, allows to record behavior, contexts, and potentially personal traits and characteristics. These metrics are increasingly used as the building blocks of digital products and services (e.g., digital health, marketing), to inform scientific discovery (e.g., behavioral science), and to provide information for the public and for decision makers (e.g., transmission tracking, traffic conditions).

This course is designed to introduce students to latest methods for the automated collection of behavioral data with mobile sensing methods. We will first cover mobile sensing methods and contrast them with more traditional approaches to data collection (e.g., surveys). Second, we will learn about some essential types of human behavior that are tracked and analyzed in many data-driven products (e.g., digital media use, communication behavior, social behavior, physical activity). Third we will cover some of the most important applications that these data could be used for (e.g., digital health, psychological assessment, targeting). Finally, we will discuss and reflect on the practical, ethical, and privacy implications that data-driven products bring along to ensure not only compliance with laws, but also acceptance of behaviorally driven products by users/customers.

Course structure and indications of the learning and teaching design

This course counts 6 credits. Accordingly, the total workload for students is 180 hours. This includes self-study, campus time, project work, discussions, and all examinations. The course is conducted in hybrid format with blended learning elements via StudyNet and synchronous class meetings. The course can also be held completely online if necessary.

Further explanation of the teaching and learning design:

This course will be facilitated online through StudyNet. Required readings and other course material (e.g., relevant video clips) will be posted on StudyNet. Announcements will also be made through the StudyNet site. You can submit any questions you have to the relevant discussion thread. Please post content-related questions in StudyNet so everyone can benefit from the answers - content related emails will not be answered by the teaching team! The course design consists of three learning pillars:

1. **Lecture Videos & Readings (asynchronous):** Short lecture videos, interviews with recognized international experts (e.g., Gabriella Harari, Stanford), and readings will be posted on StudyNet, under the Modules for each class session. You are responsible for watching and keeping up with the lecture videos throughout the semester and for completing the readings. Once you have finished watching the videos for a class session, you will take a lecture quiz as a comprehension check. The content from the videos and the readings will be helpful for the development of a behavioral application or business idea, that you and your group will present at the end of the semester. The content of the videos and readings will also be immediately relevant for the exam.
2. **Class Meetings (synchronous):** In addition to the online lecture part, our class will also meet in in-person, every week. These synchronous meetings are offered as a way for you to connect with and to work more closely with your classmates and teaching team. Our goals during the synchronous meetings are to have conversations about the course materials and for the students to collaboratively work on their project concepts. Moreover, the learning experience will also be complemented by interactive exercises and discussions. To facilitate conversations, these meetings will not be recorded nor offered in hybrid. The slides and any other shared materials will be uploaded to StudyNet and available to everyone in the course. The basis for the discussions during these meetings will be the questions posed in the StudyNet discussion thread prompts each week. But you should also feel free to bring any questions to these meetings that you would like to ask or discuss with your classmates and the teaching team.

3. **Independent Group Project (asynchronous and part of examination):** To apply the knowledge that was gained from the online materials and the class meetings, students will work on a group project throughout the semester (see description in the examination section). More details will be provided during the course.

Suggested time budget:

Content	Commitment in hours
Online videos, guest lectures, take quizzes, peer-discussions	30h
Self-study lecture content, lecture preparation & readings	65h
Group work and discussions in offline sessions	18.5h
Project work for presentation	65h

Mid-term exam	1.5h
Total	180h

Course literature

Readings will be provided for each session specifically, via StudyNet.

Additional course information

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Examination information

Examination sub part/s

1. Examination sub part (1/3)

Examination modalities

Examination type	Presentation
Responsible for organisation	decentral
Examination form	Oral examination
Examination mode	Analog
Time of examination	Term time
Examination execution	Asynchronous
Examination location	On Campus
Grading type	Group work group grade
Weighting	25%
Duration	--

Examination languages

Question language: English
Answer language: English

Remark

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Examination-aid rule

Free aids provision

Basically, students are free to choose aids. Any restrictions are defined by the faculty members in charge of the examination under supplementary aids.

Supplementary aids

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2. Examination sub part (2/3)

Examination modalities

Examination type	Written examination
Responsible for organisation	decentral
Examination form	Written exam
Examination mode	Digital
Time of examination	Term time
Examination execution	Synchronous
Examination location	On Campus
Grading type	Individual work individual grade
Weighting	50%
Duration	90 mins.

Examination languages

Question language: English
Answer language: English

Remark

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Examination-aid rule

Closed Book

The use of aids is prohibited as a matter of principle, with the exception of pocket calculator models of the Texas Instruments TI-30 series and, in case of non-language exams, bilingual dictionaries without any handwritten notes. Any other aids that are admissible must be explicitly listed by faculty members in the paragraph entitled “Supplementary aids” of the course and examination fact sheet; this list is exhaustive.

Procuring any aids, as well as ensuring their working order, is the exclusive responsibility of students.

Supplementary aids

The examination will be conducted as a digital examination with the following admissible media:

- **Private notebook** (compulsory) – no tablets!
- **Mains adapter** (compulsory)
- External computer mouse (optional)
- External computer keyboard (optional)

We would like to point out explicitly that the examination cannot be taken with tablets/iPads.
You will need the following for the examination:

- Appliance and system requirements: cf. StudentWeb
- Special software: Respondus LockDown Browser
- HSG log-in including access modalities
- WLAN – operational access to Eduroam

Please make sure that:

- all the updates have been carried out before the examination.
- In the run-up to digitally conducted examinations, technical tests (mock examinations without grades) are carried out.
- Participation is urgently required!
- You yourself are responsible for the perfect working order of your appliance, cf. also StudentWeb.
- Keyboard stickers in the language in which the examination is conducted (such as English, German, as well as Arabic,
  Chinese, Russian, Japanese, etc. in language examinations) are permitted. The stickers must be identical with the original
  characters of the language concerned.

Any use of additional appliances that are not listed above will be strictly prohibited throughout the examination.

Any necessary actions with another software or additional appliances require the express prior approval of the persons in charge of the examination and are only permitted under supervision.

Any infringement of these rules of conduct may be sanctioned as an infringement of the rules of the University

3. Examination sub part (3/3)

Examination modalities

Examination type	Written work
Responsible for organisation	decentral
Examination form	Written work
Examination mode	Digital
Time of examination	Term time
Examination execution	Asynchronous
Examination location	Off Campus
Grading type	Group work group grade
Weighting	25%
Duration	--

Examination languages

Question language: English
Answer language: English

Remark

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Examination-aid rule

Free aids provision

Basically, students are free to choose aids. Any restrictions are defined by the faculty members in charge of the examination under supplementary aids.

Supplementary aids

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Examination content

1) Exam (50%; Individual Assignment)

The final exam is an individual online exam. The exam itself is a mixture of multiple-choice questions and open questions based on the content covered in the preceding weeks (e.g., online videos, readings, discussions).

2) Project Pitch (25%; Group Assignment)

In the term project, groups of students will perform research and analysis of existing applications and businesses that build on behavioral data, summarize the current business landscape in a specific area of human behavior and propose a next generation application, product, or service. The results of the term project will be presented verbally and visually on the last day of the course. The project presentations should be given in a venture capital pitch style, be concise and clearly narrate the need for a new product/service from. The presentation will contribute 25% to the overall course grade.

3) Executive Summary (25%; Group Assignment)

In the executive summary examination part, groups will create a written report (10 pages) that describes all content from the presentation in a format that is easy to read and comprehend for decision makers. The report will be submitted at the end of the semester and feedback from the presentation should be implemented.

Guidelines:

Students will be randomly assigned to groups. Teams can request a non-mandatory pre-presentation meeting and can receive feedback after the group presentation. This meeting has no influence on the final grade. Presentation slides must be uploaded to Canvas before the presentation (pdf).

Timing: 15 min + 5 min Q&A + general discussion at the end.

Examination relevant literature

In addition to the lecture slides and the content of all in-class discussions, relevant journal articles, digital content and additional readings will be announced and made available during the course.

Please note

Please note that only this fact sheet and the examination schedule published at the time of bidding are binding and takes precedence over other information, such as information on StudyNet (Canvas), on lecturers' websites and information in lectures etc.
Any references and links to third-party content within the fact sheet are only of a supplementary, informative nature and lie outside the area of responsibility of the University of St.Gallen.

Documents and materials are only relevant for central examinations if they are available by the end of the lecture period (CW51) at the latest. In the case of centrally organised mid-term examinations, the documents and materials up to CW 43 (Freitag, 24. October 2025) are relevant for testing.

Binding nature of the fact sheets:

- Course information as well as examination date (organised centrally/decentrally) and form of examination: from bidding start in CW 34 (Thursday, 21 August 2025);
- Examination information (supplementary aids, examination contents, examination literature) for decentralised examinations: in CW 42 (Monday, 13. October 2025);
- Examination information (supplementary aids, examination contents, examination literature) for centrally organised mid-term examinations: in CW 43 (Freitag, 24. October 2025);
- Examination information (regulations on aids, examination contents, examination literature) for centrally organised examinations: two weeks before ending with de-registration period in CW 45 (Monday, 03. November 2025).